

Modbus _ RTU Communication

Description

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Communication data format

- During communication, the data is sent back in the form of word (WORD — 2 bytes). In each word sent back, the high byte is in the front and the low byte is in the back. If two words are sent back consecutively (such as long shaping), the high word is in the front and the low word is in the back.

Data type	Number of registers	Number of bytes	Explain
Character type	1	1	Send back two characters at a time, and use 0 to supplement less than two characters.
Integer	1	2	Send back once, high byte first, low byte last
Long	2	4	Send it back in two words, with the high word in the front and the low word in the back.

Frame format

• Register content query (function code 03H)

The start address and end address of the query must be the start address and end address of a complete data block. Otherwise, the data returned is incorrect. For example, if the start address of the register of the device serial number is 186 and the length is 12, the start address cannot be between 186 and 198 during the query. Similarly, the end address (start address + number of registers read) cannot fall within this range.

Frame format sent by upper computer

Byte order	Code	Examples	Explain
0	Device address	01H	Device address (1 ~ 247)
1	03H	03H	Function Code
2	Start Register Address High Byte	00H	Upper 8 bits of register address
3	Start register address low byte	10H	Lower 8 bits of register address
4	Number of registers high byte	00H	Upper 8 bits of register number
5	Low byte of register number	02H	Number of registers Lower 8 bits
6	CRC16 check high byte	C0H	High 8 bits of CRC16 check
7	CRC16 check low byte	CBH	CRC16 check lower 8 bits

The lower computer parses the returned frame format successfully.

Byte order	Code	Explain
0	Device address	Device address (1 ~ 247)
1	03H	Function Code
2	Number of bytes of data sent back	N = number of registers requested * 2
3	1st register data high byte	
4	1st register data low byte	
.....		
.....		
	Nth register data high byte	
	Nth register data low byte	
N+3	CRC16 check high byte	The bytes before the CRC are checked
N+4	CRC16 check low byte	

Lower computer analysis error return frame format

Byte order	Code	Explain
0	Device address	Device address (1 ~ 247)
1	03H	Function Code
2	Number of bytes of data sent back	N = number of registers requested * 2
3	1st 0	A total of N zeros are looped back
4	2nd 0	
.....		
.....		
	Nth-1st 0	
	Nth 0	
N+3	CRC16 check high byte	The bytes before the CRC are checked
N+4	CRC16 check low byte	

Register reading example:

Read the data from the RMS value of the mains voltage (start register 202) to the average value of the mains power, where the mains voltage returns 220.0 V, Mains frequency returns 50.0 Hz and average mains power returns 1200 w

Host computer: 01 03 00 CA 00 03 25 F5

Lower computer: 01 03 06 08 FC 13 88 04 B0 F7 F3

• Register content setting (function code 10 H)

Frame format sent by upper computer

Byte order	Code	Examples	Explain
0	Device address	01H	Device address (1 ~ 247)
1	10H	10H	Function Code
2	Start Register Address High Byte	01H	Upper 8 bits of register address
3	Start register address low byte	10H	Lower 8 bits of register address
4	Number of registers high byte	00H	Upper 8 bits of the number of registers (constantly equal to 0)
5	Low byte of register number	02H	Number of registers Lower 8 bits
6	Number of bytes to be written		N = Number of registers * 2
7	1st register data high byte		
8	1st register data low byte		
.....			
.....			
	Nth register data high byte		
	Nth register data low byte		
N+7	CRC16 check high byte		High 8 bits of CRC16 check
N+8	CRC16 check low byte		CRC16 check lower 8 bits

The lower computer parses the returned frame format successfully.

Byte order	Code	Examples	Explain
0	Device address	01H	Device address (1 ~ 247)
1	10H	10H	Function Code
2	Start Register Address High Byte	01H	Upper 8 bits of register address
3	Start register address low byte	10H	Lower 8 bits of register address
4	Number of registers high byte	00H	Upper 8 bits of the number of registers (constantly equal to 0)
5	Low byte of register number	02H	Number of registers Lower 8 bits
6	CRC16 check high byte	41H	High 8 bits of CRC16 check
7	CRC16 check low byte	F1H	CRC16 check lower 8 bits

Lower computer analysis error return frame format

Byte order	Code	Explain
0	Device address	Device address (1 ~ 247)
1	90H	Function Code
2	Error code	Error code
3	CRC16 check high byte	High 8 bits of CRC16 check
4	CRC16 check low byte	CRC16 check lower 8 bits

Error code description

Code	Explain
01H	Read-only register operated on
03H	Write data that exceeds the acceptable range
07H	Registers that are not allowed to be modified in the current operating mode

Register write example:

Set the output voltage (start register 320) to 220V

Host computer: 01 10 01 40 00 01 02 08 98 BE 3A

Lower computer: 01 10 01 40 00 01 01 E1

Device register address

- ◆ **R:** Indicates that the 03 H command is supported if it is readable. **W:** means writable, that is, supports 10 H commands. **Int:** Integer; **Long:** Long; **UInt:** unsigned integer; **ULong:** unsigned long integer; **ASC:** ASCII code **Max:** maximum value; **Min:** minimum value
- ◆
- ◆ Addresses in the following table are in decimal notation

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Equipment fault code		ULong	100	2	R	32-bit fault code. Each bit corresponds to a fault code. See the fault code table for details. Fault code 1 corresponds to bit1, fault code 2 corresponds to bit2, and so on.
Reserved			102	2		Reserved address
Obtain the warning code for unmasked processing			104	2		The 32-bit warning code is described in the warning code description
Reserved			106	2		Reserved address
Obtain the warning code after shield processing		ULong	108	2	R/W	The 32-bit warning code is described in the warning code description
Reserved			110	61		Reserved address
Device type		UInt	171	1	R	
Device name		ASC	172	12	R/W	Device name, written or read in ASCII
Invalid data		UInt	184	1	R	Agreement number, return 1 for this agreement
Reserved			185	1		Reserved address
Device serial number		ASC	186	12	R	
Reserved			198	2		Reserved address
Invalid data		UInt	200	1		Internal command

Working mode		UInt	201	1	R	0: Power on mode 1: Standby mode 2: Mains mode 3: Off-grid mode 4: Bypass mode 5: Charging mode 6: Failure Mode
Mains voltage effective value	0.1v	Int	202	1	R	
Mains frequency	0.01Hz	Int	203	1	R	
Average mains power	1w	Int	204	1	R	

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Effective value of inverter voltage	0.1v	Int	205	1	R	
Effective value of inverter current	0.1A	Int	206	1	R	
Inverter frequency	0.01Hz	Int	207	1	R	
Inverter power average	1w	Int	208	1	R	Positive indicates inverter output and negative indicates inverter input
Inverter charging power	1w	Int	209	1	R	
Effective value of output voltage	0.1v	Int	210	1	R	
Effective value of output current	0.1A	Int	211	1	R	
Output frequency	0.01Hz	Int	212	1	R	
Output active power	1w	Int	213	1	R	
Output apparent power	1VA	Int	214	1	R	
Average battery voltage	0.1v	Int	215	1	R	
Average battery current	0.1A	Int	216	1	R	

Average battery power	1w	Int	217	1	R	
Invalid data			218	1		Internal command
Average PV voltage	0.1v	Int	219	1	R	
Average PV current	0.1A	Int	220	1	R	
Reserved			221	2		Reserved address
Average PV power	1w	Int	223	1	R	
Average PV charging power	1w	Int	224	1	R	

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Percent of load	1%	Int	225	1	R	
DCDC temperature	1°C	Int	226	1	R	
Inverter temperature	1°C	Int	227	1	R	
Reserved			228	1		Reserved address
Battery percentage	1%	UInt	229	1	R	
Invalid data			230	1		Internal command
Power flow status		UInt	231	1	R	See the description of power flow flag bit for details.
Battery current filter average	0.1A	Int	232	1	R	A positive number indicates charging and a negative number indicates discharging.
Average value of inverter charging current	0.1A	Int	233	1	R	
Average PV charging current	0.1A	Int	234	1	R	
Invalid data			235	1		Internal command
Invalid data			236	1		Internal command
Reserved			237	63		Reserved address
Output mode		UInt	300	1	R/W	0: single machine; 1: parallel; 2: Three-phase combination-P1 3: Three-phase combination-P2 4: Three-phase combination-P3

Output priority		Uint	301	1	R/W	0: Main-PV-Battery (UTI) 1: PV-mains-battery (SOL) [priority inverter] 2: PV-battery-mains (SBU) 3: PV-Mains-Battery (SUB) [Priority Mains]
Input voltage range		Uint	302	1	R/W	0: APL; 1: UPS;
Buzzer mode		Uint	303	1	R/W	0: mute in all cases; 1: Sound when the input source changes or there is a specific warning or fault; 2: Sound when there is a specific warning or fault; 3: Sound in case of fault;

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Reserved			304	1	R/W	Reserved address
LCD backlight		Uint	305	1	R/W	0: Timed closing; 1: Always on;
LCD automatically returns to the home page		Uint	306	1	R/W	0: Do not return automatically; 1: Automatic return after 1 minute;
Energy saving mode switch		Uint	307	1	R/W	0: Energy saving mode off; 1: Energy-saving mode on;
Overload automatic restart		Uint	308	1	R/W	0: Overload failure does not restart; 1: Automatic restart in case of overload;
Over-temperature automatic restart		Uint	309	1	R/W	0: No restart for over-temperature fault; 1: Automatic restart for over-temperature fault;
Overload Bypass Enable		Uint	310	1	R/W	0: forbidden; 1: Enable;
Reserved			311	2		Reserved address
Battery Eq Mode Enable		Uint	313	1	R/W	0: forbidden; 1: Enable;
Warning Mask [!]		ULong	314	2	R/W	The warning corresponding to 1 is normally displayed, and the warning corresponding to 0 is shielded.
Dry contact		Uint	316	1	R/W	0: normal mode; 1: Grounding box mode;
Reserved			317	3		Reserved address

Output voltage	0.1v	Uint	320	1	R/W	2200: 220V output; 2300: 230v output; 2400: 240v output;
Output frequency	0.01Hz	Uint	321	1	R/W	5000: 50Hz output; 6000: 60Hz output;
Battery type		Uint	322	1	R/W	0: AGM; 1: FLD; 2: USER; 3: Li1 4: Li2 5: Li3 6: Li4
Battery overvoltage protection point [A]	0.1v	Uint	323	1	R/W	Range: (B + 1V * J) ~ 16.5v * J

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Maximum charge voltage [B]	0.1v	Uint	324	1	R/W	Range: C ~ (A-1v)
Floating charge voltage [C]	0.1v	Uint	325	1	R/W	Range: (12v * J) ~ B
Mains mode battery discharge recovery point [D]	0.1v	Uint	326	1	R/W	Range: (B-0.5V * J) ~ Max (12V * J, E) Set to 0 to indicate a full recovery
Battery low voltage protection point in mains mode [E]	0.1v	Uint	327	1	R/W	Range: Min (14.3v * J, D) ~ Max (11v * J, F)
Reserved			328	1		Reserved address
Off-grid mode battery low voltage protection point [F]	0.1v	Uint	329	1	R/W	Range: (10v * J) ~ Min (13.5v * J, E)
Waiting time from constant voltage to floating charge	min	Uint	330	1	R/W	Range: 1 ~ 900 min Set to 0 to default to 10 min
Battery charging priority		Uint	331	1	R/W	0: mains supply is preferred; 1: PV priority; 2: PV is at the same level as mains supply; 3: PV charging only allowed

Maximum charge current [G]	0.1A	Uint	332	1	R/W	Range: Max (10 A, H) ~ 80 A
Maximum mains charging current [H]	0.1A	Uint	333	1	R/W	Range: 2A ~ G
The charging voltage of Eq	0.1v	Uint	334	1	R/W	Range: C ~ (A-0.5v * J) http://c/
bat_eq_time	min	Uint	335	1	R/W	Range: 0 ~ 900
Eq timed out	min	Uint	336	1	R/W	Range: 0 ~ 900
Two-time Eq charge interval	day	Uint	337	1	R/W	Range: 1 ~ 90
Automatic Mains Output Enable		Uint	338	1	R/W	0: No mains power output without pressing the power button 1: Automatic mains power output without pressing the power button
Reserved			339	2		Reserved address

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Mains mode battery discharge SOC protection value [K]	1%	Uint	341	1	R/W	Range: 20% ~ 50%
Mains mode battery discharge SOC recovery value	1%	Uint	342	1	R/W	Range: 60% ~ 100%
Battery discharge SOC protection value in off-grid mode	1%	Uint	343	1	R/W	Range: 3% ~ Min (K, 30%)
Reserved			344	7		
Maximum discharge current protection	1A	Uint	351	1	R/W	Maximum discharge current protection value in stand-alone mode
Reserved			352	54		
Boot mode		Uint	406	1	R/W	0: Can be powered on locally or remotely 1: Only local boot 2: Remote boot only
Reserved			407	13		Reserved address
Remote switch		Uint	420	1	R/W	0: Remote shutdown 1: Remote power-on
Invalid data			421	1		Internal command
Reserved			422	3		Reserved address
Forcing the charge of Eq		Uint	425		W	1: Manually force Eq to charge once

Exits the fail-locked state		Uint	426		W	1: Exit the fault lock state (it will take effect only when the machine enters the fault mode)
Invalid data			427	1		Internal command
Reserved			428	22		Reserved address
Invalid data			450	7		Internal command
Reserved			457	3		Reserved address
Clear the record			460	1	W	0 xAA: clear the operation record and fault record (effective in non-off-grid mode)
Reset user parameters			461	1	W	0 xAA: User parameters are restored to default values (works in non-off-grid mode)

Data name	Unit	Data format	Start address	Number of registers	Read and write	Remark
Invalid data			462	6		Internal command
Reserved			468	32		Reserved address
Invalid data			500	34		Internal command
Reserved			534	66		Reserved address
Invalid data			600	34		Internal command
Program version		ASC	626	8	R	
Reserved			634	7		Reserved address
Rated power	w	UInt	643	1	R	
Rated number of cells [J]	PCS	UInt	644	1	R	
Reserved			645	55		Reserved address
Fault record storage information [K]		ULong	700	2	R	Upper 16 bits: the position of the latest record; Lower 16 bits: total number of existing fault messages
Fault Information Query Index		UInt	702	1	R/W	Set the fault information index to be queried, range: 0 ~ total number of existing fault information
Fault Record [M]			703	26	R	See the fault record format for details.
Run the log			729	16	R	See the operation log description for details
Reserved			745	5		Reserved address

Fault code table

Fault code	Explain
bit1	Inverter overtemperature
bit2	DCDC overtemperature
bit3	Battery overvoltage
bit4	PV overtemperature
bit5	The output is shorted
bit6	Inverter overvoltage
bit7	Output overload
bit8	Busbar overvoltage
bit9	Bus soft start timeout
bit10	PV overcurrent
bit11	PV overpressure
bit12	Battery overcurrent
bit13	Inverter overcurrent
bit14	Busbar low voltage
bit15	Reserved
bit16	Inverter DC component is too high
bit17	Reserved
bit18	Output current zero bias is too large
bit19	Inverter current zero bias is too large
bit20	Excessive zero bias of battery current
bit21	PV current zero bias is too large
bit22	Inverter low voltage
bit23	Inverter negative power protection
bit24	Loss of parallel system host
bit25	Abnormal synchronization signal of parallel system
bit26	Incompatible battery type
bit27	Incompatible parallel version

Warning code description

A system warning is a 32-bit unsigned long integer, one for each bit, Each bit can also be masked by a warning mask I. After masking, the corresponding warning will not be read on the LCD or by a command.

• Warning code table

Warning Code	Explain
bit 0	Mains supply zero-crossing loss
bit 1	Mains waveform is abnormal
bit 2	Mains overvoltage
bit 3	Mains undervoltage
bit 4	The mains supply is too frequent
bit 5	Mains underfrequency
bit 6	PV undervoltage
bit 7	Overtemperature
bit 8	Low battery voltage
bit 9	Battery is not connected
bit 10	Overload
bit 11	Charge the battery Eq
bit 12	The battery is discharged to a low voltage and has not been charged back to the recovery point
bit 13	Output power derating
bit 14	The fan is blocked
bit 15	PV energy too low to use
bit 16	Parallel communication is interrupted
bit 17	Inconsistent single parallel output mode
bit 18	Excessive voltage difference of parallel battery
bit 19	Abnormal lithium battery communication
bit 20	Battery discharge current exceeds the set value
bit 21~31	Reserved

Description of power flow signs

The power flow has a 16-bit unsigned integer representation, and the meaning of each bit is as follows:

• Power flow level table

Bit	Explain
bit0~1	0: PV is not connected to the system 1: PV is connected to the system
bit2~3	0: The mains supply is not connected to the system 1: The mains supply has been connected to the system
bit4~5	0: The battery will not be charged or discharged 1: Battery charging 2: Battery discharge
bit6~7	0: The system does not output on-load 1: System output with load
bit8	0: Mains power is not charged 1: Mains charging
bit9	0: PV not charged 1: PV charging
bit10	0: Battery icon is on 1: Battery icon off
bit11	0: PV icon is on 1: PV icon off
bit12	0: The electric map of the city lights up 1: Mains icon off
bit13	0: The load icon is on 1: Load icon off

CRC check algorithm

```

0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40
} ;

```

```

const char auchCRCLo[] = {
0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06,
0x07, 0xC7, 0x05, 0xC5, 0xC4, 0x04, 0xCC, 0x0C, 0x0D, 0xCD,
0x0F, 0xCF, 0xCE, 0x0E, 0x0A, 0xCA, 0xCB, 0x0B, 0xC9, 0x09,
0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9, 0x1B, 0xDB, 0xDA, 0x1A,
0x1E, 0xDE, 0xDF, 0x1F, 0xDD, 0x1D, 0x1C, 0xDC, 0x14, 0xD4,
0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3,
0xF2, 0x32, 0x36, 0xF6, 0xF7, 0x37, 0xF5, 0x35, 0x34, 0xF4,
0x3C, 0xFC, 0xFD, 0x3D, 0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A,
0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38, 0x28, 0xE8, 0xE9, 0x29,
0xEB, 0x2B, 0x2A, 0xEA, 0xEE, 0x2E, 0x2F, 0xEF, 0x2D, 0xED,
0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60,
0x61, 0xA1, 0x63, 0xA3, 0xA2, 0x62, 0x66, 0xA6, 0xA7, 0x67,
0xA5, 0x65, 0x64, 0xA4, 0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F,
0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB, 0x69, 0xA9, 0xA8, 0x68,
0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA, 0xBE, 0x7E,
0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71,
0x70, 0xB0, 0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92,
0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C,
0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B,
0x99, 0x59, 0x58, 0x98, 0x88, 0x48, 0x49, 0x89, 0x4B, 0x8B,
0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42,
0x43, 0x83, 0x41, 0x81, 0x80, 0x40
} ;

```

```

unsigned short sModbusCrc16(INT8U *chMsg, INT16U dataLen)
{

```

```
unsigned char ubCRCHi = 0xFF;
unsigned char ubCRCLo = 0xFF;
unsigned char duwIndex; while
(dataLen --)
{
    duwIndex = 0xff&(ubCRCHi ^ *chMsg++);
    ubCRCHi = 0xff&(ubCRCLo ^ auchCRCHi[duwIndex]);
    ubCRCLo = auchCRCLo[duwIndex];
}
return (ubCRCHi << 8 | ubCRCLo);
```


